

11. Wellbore Seismic

Summary

For various purposes petrophysicists often work together with geophysicists and interpreters in the different stages of a life-cycle of a field.:

- o Generation of a time-to-depth conversion curve, to convert seismic horizons from two-way travel time to depth (Z-T curves) and/or to convert logs (in depth) to time. This is normally done using checkshot data combined with sonic log data. Checkshots use surface acoustic sources and downhole geophones to measure integrated sound velocities at specific depth points. The sonic is used to interpolate between such discrete points.
- o Generation of synthetic seismograms: using the sonic and the density logs a reflectivity curve can be generated. Filtering this curve down to seismic frequencies gives a log derived seismic display which can be compared to the real (surface) seismic, e.g. for modelling purposes, quality control, etc.
- o The influence of the pore fluid type on the seismic can be modelled using the Gassmann equation.
- o The generation of trendcurves

These topics are dealt with in the current chapter.

References

- Nolen – Hoeksema, Richard C.; *Modulus – porosity relations, Gassmann's equations, and the low – frequency elastic – wave response to fluids*. Geophysics, Vol 65 no 5 (september – october 2000), p.1355 – 1363.
- Brie, A. et al.; *Shear Sonic Interpretation in Gas – Bearing Sands*. SPE 30595 (SPE Annual Technical Conference, Dallas 22 – 25 October 1995).
- Serra, O.; *Fundamentals of Well Log Interpretation*. Elsevier 1984, p197
- *Rock Properties Manual* by Jacques Staal
- *AVF discussion on Gassmann in Logic* (SKS, Petrophysics)
- *AVF discussion on invaded zone saturations for oil-based mud* (SKS, Petrophysics)

Petrophysics and Seismic

Petrophysical logs are acquired in the borehole and investigate typically an area of one to several feet away from the borehole. Hence, their depth of investigation is extremely limited compared to seismic (kilometers). On the other hand log resolution can range from metres (resistivity tools) down to centimetres (borehole imaging tools), which compares favourably to seismic resolution (typically 15 metres, at best 5 metres): see *Figure 1*.

Typical petrophysical items related to seismic are:

- Generation of a time-to-depth conversion curve, using checkshots.
- Generation of synthetic seismograms.
- Assessing the influence of the pore fluid type on the seismic using the Gassmann equation.

Checkshots

In a checkshot survey, a surface acoustic source (e.g. air-gun) is used in combination with downhole geophones to measure integrated sound velocities at specific depth points. Hence, the one-way travel time is measured (from surface source to borehole geophone). This time has to be multiplied by a factor of two to compare it to the two-way travel time measured by seismic.

Apart from measuring the first arrival transit time only, the checkshot survey can also be used to analyse the full wave train. This is called Vertical Seismic Profiling (VSP): it can be analysed in terms of up and down going (acoustic) events and thus yield a seismic-like image. VSP's will not be discussed here.

In order to compare checkshot data to sonic log and/or seismic data several factors have to be accounted for:

- The difference between seismic reference datum (where time = 0 ms and normally sea level), ground level (or rather hydrophone level from which the checkshot is measured) and the Derrick Floor Elevation (from which the logs are measured).
- The difference between True Vertical Depth Sub Sea (seismic & checkshot) and Depth Along Hole Below Derrick Floor (logging depth). Moreover, if the well is deviated, the checkshot source should preferably be perpendicular above the downhole geophone, and hence the source has to move ("walk away") when the geophone location is changed. Otherwise a correction has to be made for the non-verticality of the acoustic waves going from source to geophone.
- Both the sonic and the checkshot measurement should be checked and corrected errors. Old sonic logs often show "cycle skipping" and "noise peaks" (see *Figure 2*); these spikes have to be edited out. Typical "*basic or mechanical edits*" to be done to the sonic are: depth matching, joining of curves, removal of casing shoes effects and removal of cycle skips & noise peaks. Other means of editing are "*Interpretive editing*" (using trend curves or inter-log relationships) or *modelling* (e.g. using Gassmann: see further below).

The Z-T (Depth-Time) relationship

Once the checkshot and sonic log data have been edited and corrected satisfactorily, the semi-continuous sonic log (typically 2 samples per foot) can be used to interpolate the discrete irregular sampled checkshot data points: see *Figure 3*.

However, the sonic data might differ from the checkshot data. The relative difference between the two way (seismic) time determined from the check shots and the sonic log is referred to as **drift**. Drift is often caused by alteration of the borehole wall during drilling and is normally most noticeable in shale sections. This alteration results in integrated sonic times being longer than the check shot times and hence is termed **negative drift**. Other reasons for negative drift may be cycle skipping & large borehole effects (*Figure 4*), or non-vertical alignment of the checkshot source/geophone system. **Positive drift** is mainly attributed to frequency effects i.e. higher frequency sonic waves (20 - 30 KHz) travel faster than the lower frequency seismic waves (60 Hz). In deviated wells, lateral velocity changes can also account for positive drift.

The drift curve is mainly used to isolate anomalous checkshots, e.g. due to erroneous picking of first breaks (first arrival), reflections and multiples (multiple reflections) close to formation boundaries or checkshots having spacings less than 200 ft. Once all errors have been removed, the sonic can be force fitted through the checkshot points.

Synthetics

Application

From sonic and density logs a synthetic seismogram can be constructed (see below) that can be used:

- as a quality control on the actual surface seismic data
- study the impact of seismic processing, e.g. use of zero phasing (see below), justification of pick criteria, etc.
- for modelling studies, e.g. calculating the sensitivity of the seismic to changes in fluid type, fluid content, etc.

Impedance and Reflection Coefficient

Acoustic waves travelling through the sub-surface will be reflected by and/or transmitted through a lithological boundary depending on the difference in acoustic impedance of the adjacent layers.

Acoustic impedance (Z) is defined as:

$$\text{Ln}(\text{density} * \text{velocity}) \quad 1)$$

It is standard practice to **normalise the impedance** w.r.t. water velocity (+/- 4921 ft/sec):

$$\text{Ln}((\text{density} * \text{velocity})/4921)) \quad 2)$$

If the sonic data is in $\mu\text{sec/m}$ the water velocity is 1500 m/sec.

The reflection coefficient (R) between two layers is defined as:

$$R = (\rho_2 V_2 - \rho_1 V_1) / (\rho_2 V_2 + \rho_1 V_1) \quad 3)$$

where ρ and V are the density and propagation velocity of a given layer. Subscript 1 denotes the top layer and subscript 2 the bottom layer. From the formula, it can be seen that in the case of an increase in impedance ($\rho_2 V_2 > \rho_1 V_1$) R is positive.

As $(\rho_2 V_2 - \rho_1 V_1)$ is usually small compared with either $\rho_2 V_2$ or $\rho_1 V_1$, R can be closely approximated by:

$$R \cong \frac{\Delta \rho V}{2 \rho V} \cong \frac{\Delta \ln(\rho V)}{2} = \frac{\Delta Z}{2} \quad 4)$$

Conventionally (and in many seismic applications) the computation of reflection coefficient (R) from log data is taken as half the difference in normalised acoustic impedance between successive intervals.

The reflection coefficient is subsequently used to compute filtered time synthetics for comparison with seismic data in a reflectivity section.

Seismic phase

The seismic phase describes the shape of the seismic wavelet. Seismic sources are **minimum phase**, i.e. the wavelet starts at time zero. However, the seismic interpretation is much easier if the loops are identified by looking at the maximum amplitude. Taking the maximum amplitude (peak) position as time zero is called **zero-phasing**, which means that the seismic has to be processed to zero-phase.

Polarity Conventions of Seismic and Well Synthetics

When surface seismic traces are being compared with well synthetics, polarity can be a source of major confusion. It is essential that the convention used in the production of a well synthetic is fully understood by both the petrophysicist and the seismic interpreter.

As discussed in the previous section, an increase in acoustic impedance results in a positive reflection coefficient. A seismic wave reflected from a low to higher impedance interface (**hard kick**) will have the same polarity as the incident wave. If the seismic source is an explosive device (normal situation), the first detected motion will be a compression event. In both land and marine seismic, the Society of Exploration Geophysicists (**SEG**) have standardised on the **SEG Acquisition Convention** that an upward motion detected by a geophone or a pressure increase detected by a hydrophone will be recorded as a negative number. Such a motion or pressure increase is the direct result of a compressional event.

The **SEG Acquisition Convention** for plotting zero phase reflectivity seismic traces is for negative values to be displayed to the left and positive values to the right, with black shading between the zero line and the positive amplitude peak. Therefore:

*the reflectivity trace of a seismic wave reflected from a layer of **increased impedance (hard kick)** will be displayed on a zero phase reflectivity seismic*

trace as a white negative loop centred at the layer interface with adjacent pre- and post-cursors as black positive peaks.

The reflectivity trace for reflection from a layer of **decreased impedance (soft kick)** will be the opposite. The acoustic impedance, reflection coefficient, recorded reflectivity and seismic trace display for both cases are shown in *Figure 5*.

Note that the above illustration is the **SEG Acquisition Convention**. For plotting seismic traces and well synthetics there is also in the industry a **SEG Positive Polarity Plotting Convention** which is totally opposite to that described above i.e. a hard kick on a zero phase reflectivity seismic trace is plotted as a black loop!

Confused ?? I can imagine but do not worry: the polarity convention used will normally be indicated on the seismic trace print and if in doubt ask the seismic processing "guru's" for clarification.

Filtered Time Synthetics

In order to compare the synthetics determined from log data with seismic data, the synthetic has to be convolved with a filter that mimicks the filtering effect that the earth layers have on seismic waves. The **Butterworth -3 dB** and **Butterworth Trapezium** (*Figure 6*) are the most commonly used. The required filters and filter parameters should be discussed with the seismic interpreter. Both filters are used to define a trapezium shaped boundary on an amplitude/frequency crossplot. Any data outside this boundary (low and high frequencies) will be excluded.

In the Butterworth -3 dB, the filter specifications are expressed as two anchor points at -3 dB (points B and C of the trapezium shape) with the slopes of the two sides of the trapezium being defined by slope order expressed in dB/octave divided by 6. The spectrum length will be usually either 1024 or 512 and zero phase with no amplitude attenuation should normally be selected.

In the Butterworth trapezium, the filter is specified by defining four corner points (points A, B, C and D). Filter length, phase and normalisation should be set as for the -3 dB filter.

A summary of the steps involved in synthetic generation is given in *Figure 7*. When the synthetics are displayed on a plot, both impedance and reflectivity should be shown to ensure that there is no confusion about the polarity.

An example of a well synthetic (and seismic-to-well match) is shown in *Figure 8*.

Gassmann Fluid Replacement

Gassmann's fluid replacement algorithm has been used since around 1970 to model the effect on seismic response of changes in porosity or fluid content in porous rock. By modelling the subsurface as a packing of rigid sphere's connected by springs and employing the parameters used to characterise elastic media, Gassmann related the compressional and shear velocities of sound in the rock to the bulk moduli of the rock grains, the empty matrix and the fluid in the pore space. The resultant equations contain a number of unknown quantities such as the bulk modulus and Poisson's ratio of the porous rock matrix, but if these parameters can be estimated in a rock of known pore fill then they can be used to predict the effect of changes in pore fill on the sonic velocity and thus ultimately on impedance and seismic response.

Application

The Gassmann equations are used to:

- model changes in pore fill and thus predict seismic response away from the wellbore on the basis of sonic and density log data. The Figure below illustrates the effect of changes in the pore gas saturation on the compressional fluid velocity.
- remove the effects of mudfiltrate invasion on sonic & density logs during initial processing.

Principle

The compressional velocity V_p depends on:

- porosity and density: these are obtained from the density log
- Poisson's ratio of the dry rock: this is normally obtained from correlation Tables in the literature
- elastic moduli of the solid (obtained from correlation Tables), the fluid (obtained from Tables and/or from Woods' mixing law [inverse modulus is linear volumetric average of constituents]) and the dry rock (see below).

Here, the bulk modulus of the dry rock is the only real unknown. It is obtained by measuring V_p in the water zone and applying the Gassmann equation to yield this bulk modulus of the dry rock. Subsequently, all parameters now being known, V_p can be calculated for other values of the water saturation, i.e. in oil or gas bearing rock. The effect will be greatest if 100% water is replaced by 95% water and 5% gas: see *Figure 9*.

The processing steps are as follows:

Step 1: For the initial fluid saturation conditions, calculate the porosity, bulk modulus and shear modulus of the fluid-filled rock.

Step 2: Solve the Gassmann equation for the bulk modulus of the dry rock.

Step 3: Calculate bulk density and fluid bulk moduli for the final fluid saturation conditions.

Step 4: Solve the Gassmann equation for the compressional sonic velocity at the final fluid saturation conditions.

An example of the Gassmann replacement, applied on the density and the sonic, is shown in *Figure 10*.

Appendix 1 provides more detail on the Gassmann substitution algorithm.

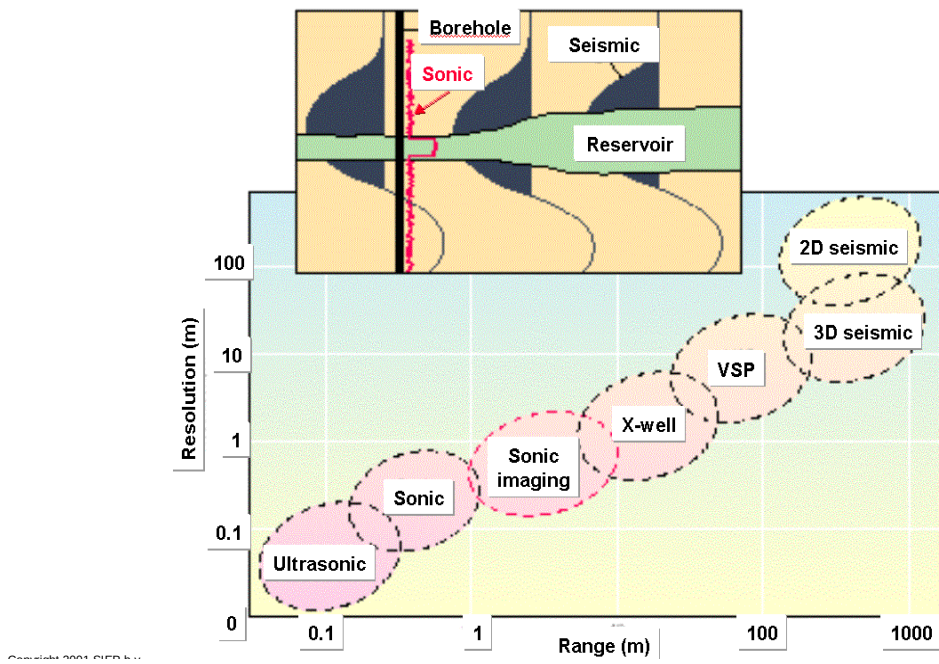


Figure 1 - Logs and Seismic...a resolution issue

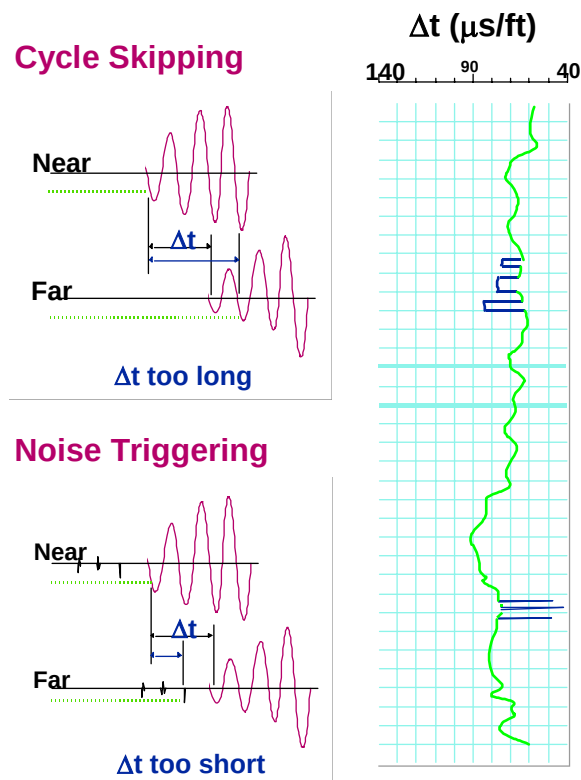


Figure 2 – Noise & Cycle Skipping

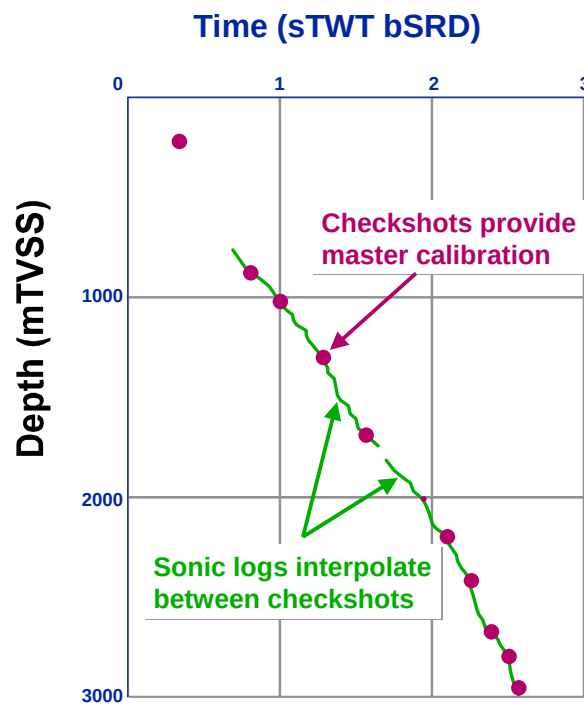


Figure 3 - Time / Depth Relationship

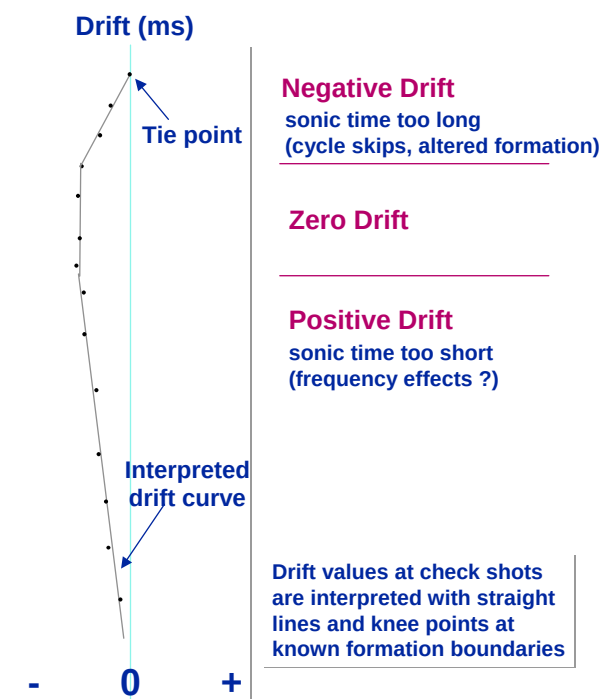


Figure 4 - Drift Curves

SEG Acquisition Convention for Seismic Traces

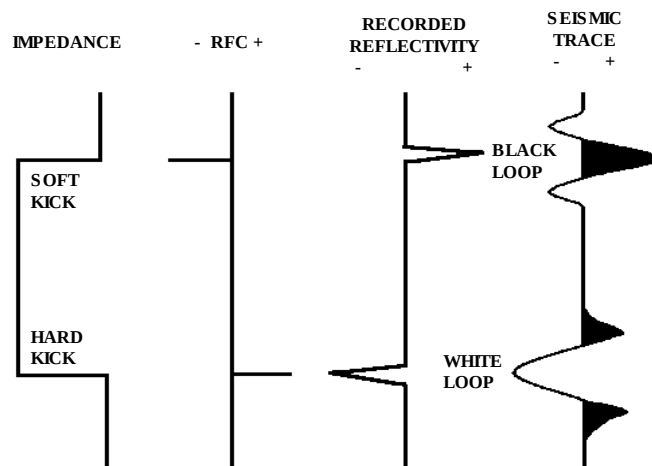
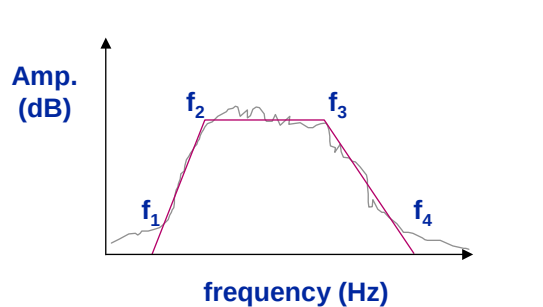


Figure 5 - Conventions for Polarity

Butterworth Filters

Method 1 Trapezium Corner Frequencies



Method 2 Slope Order and -3dB Points

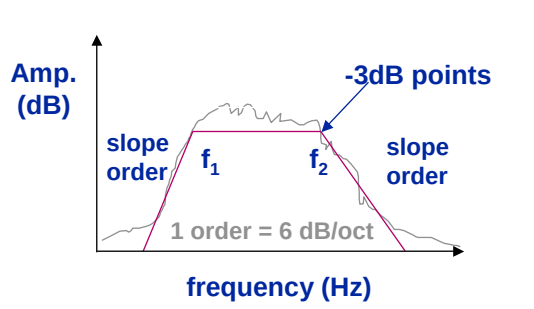


Figure 6 - Filter definitions

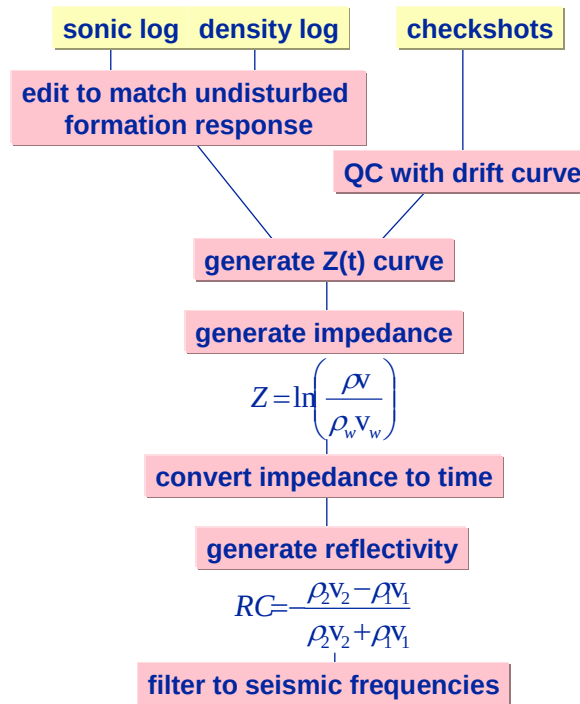


Figure 7 - The Synthetic Generation Process

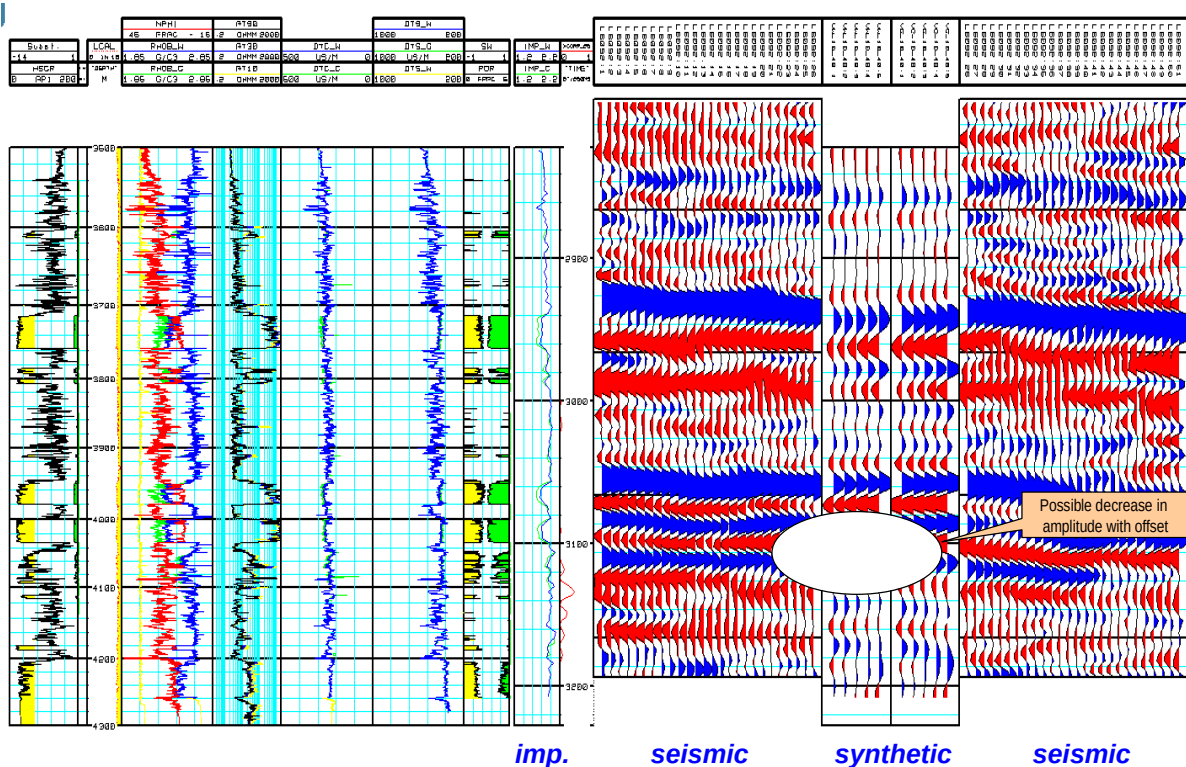
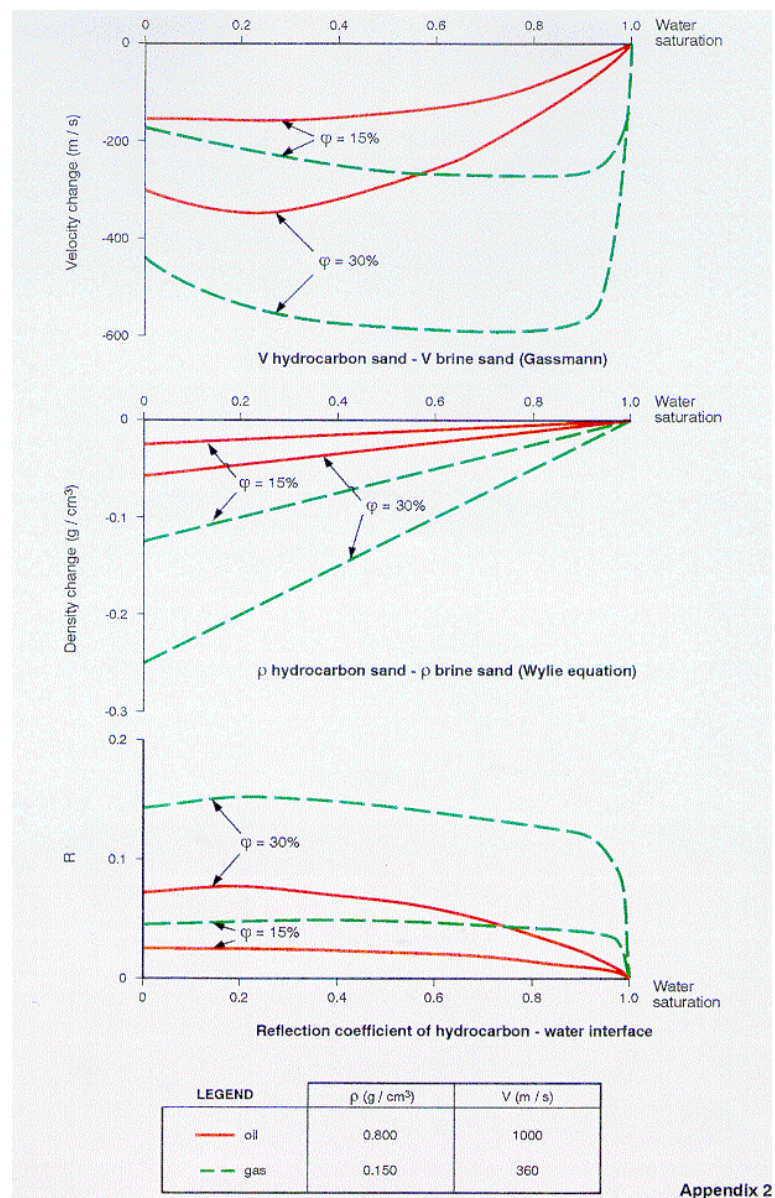


Figure 8 - An example of well synthetics spliced in a seismic section



Appendix 2

Figure 9 - The Effect of Porefill on V_p and ρ

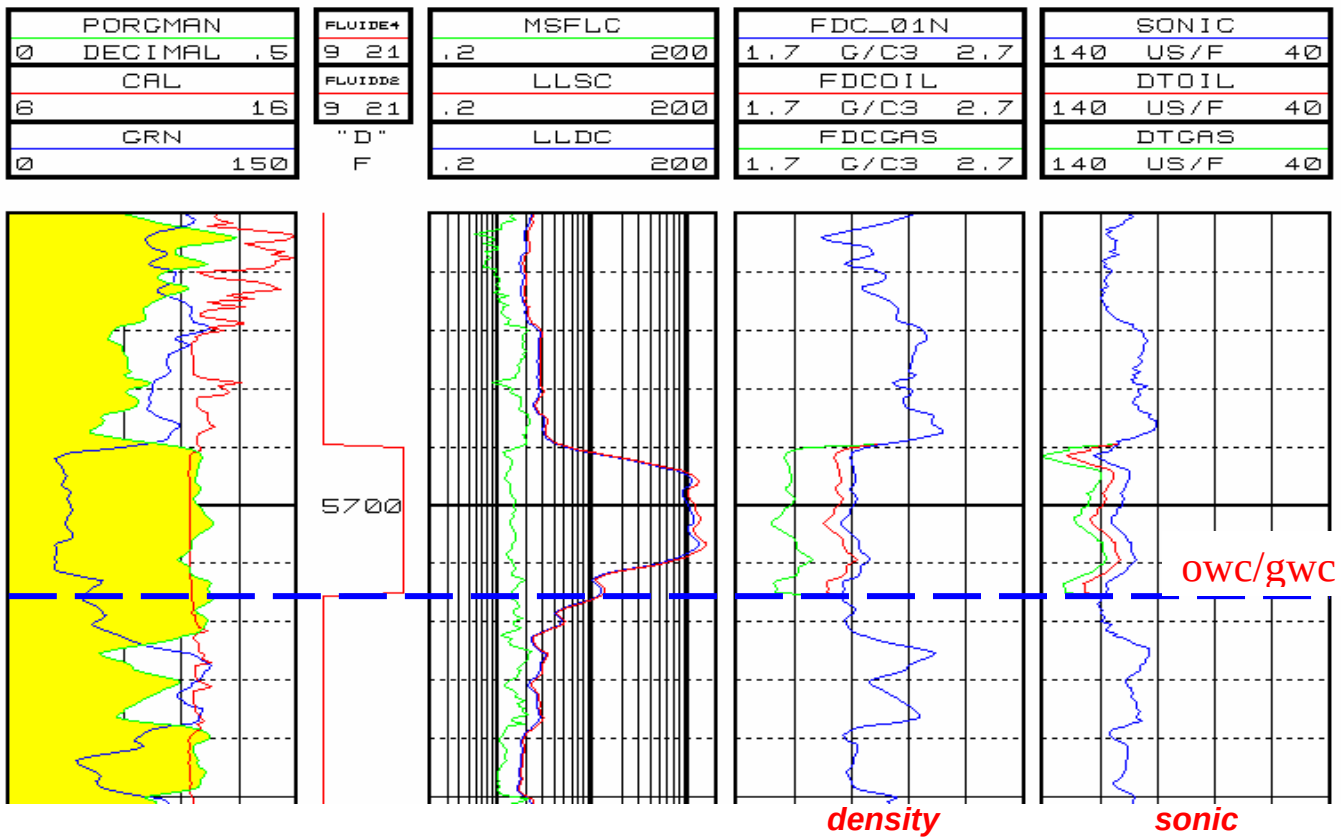


Figure 10 - The use of the Gassmann replacement algorithm: the invaded zone has been replaced by oil and gas respectively. Both the density and sonic show a more "softer" response